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Social Trust, Institution, and Economic Growth: Evidence from China

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ABSTRACT: Recent research has demonstrated the important role of social trust in economic growth. As a form of informal institution, social trust and formal institution are inextricable and intrinsically related. This article aims to investigate the relationship between social trust and institution, and their combined effects on economic growth. In an empirical investigation of cross-provincial data in the period 2001–2009 in China, our estimates suggest that the increase in social trust significantly promotes economic growth. The improvement of formal institution is also beneficial to economic growth. The effect of social trust on economic growth depends on the institution level, and this effect weakens with institutional strength.

KEY WORDS: economic growth, institution, social trust

China's economy has been developing at a faster rate since the reform and opening up policy in the 1980s. In the process, however, different regions in China show great differences in economic growth and income level. For example, the gross domestic product (GDP) per capita was 13,886.13, 5980.85, and 5071.40 in RenMinBi (RMB) terms in the east, central, and west of China, respectively, in the year of 2000. The number in the east was 2.32 times that in the central and 2.74 times that in the west. By the end of 2014, the GDP per capita in the east had reached 71,417.61 and was 1.80 times that in the central and 1.85 times that in the west, as shown in [Table 1](#). Although the differences between regions are narrowing, they are still remarkable.

To explain the above differences in growth, the classical economic theories identified land, labor, and physical capital as three basic factors. In the 1960s, neoclassical economists proposed the notion of human capital, whose importance is that it determined how productively the traditional factors could be utilized. However, recent studies show that even with the most advanced physical and human capital, it is difficult to achieve economic growth, unless individuals are willing to combine their particular skills and resources in a spirit of trust, cooperation, and commitment to common objectives.

More and more scholars began paying attention to the important role of social trust in economic growth since the mid- and late 20th century. Arrow (1972) pointed out that virtually every commercial transaction has within itself an element of trust, certainly any transaction conducted over a period of time. It can be plausibly argued that much of the economic backwardness in the world can be explained by the lack of mutual confidence. Putnam, Leonardi, and Nanetti (1993) attributed the difference in economic growth between the south and north regions of Italy to the difference in social structure and social trust. He confirmed the important role of social capital, especially social trust, in economic growth for the first time in history. Knack and Keefer (1997) presented evidence that social capital matters for measurable economic performance, using indicators of trust and civic norms from *World Values Survey* (WVS) for a sample of 29 market economies. Zak and Knack (2001) established a general equilibrium growth model in which heterogeneous agents transact and face a moral hazard problem and found that high trust societies exhibit higher rates of investment and growth by reducing transaction costs. In a high-trust environment, investors pay less time and effort in verifying the

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Table 1. GDP per capita in China: 2000–2014.

	GDP per capita (in RMB terms)			Comparison with the East		
	2000	2010	2014	2000	2010	2014
East	13,886.13	49,104.10	71,417.61	—	—	—
Center	5980.85	25,389.68	39,685.32	0.431	0.517	0.556
West	5071.40	23,451.36	38,657.31	0.365	0.478	0.541

Note: (1) GDP per capita is denominated to the price of the year 2000. (2) Source: China Statistical Yearbook.

honesty of their agents and therefore end up with higher output. Berggren, Elinder, and Jordahl (2008) conducted an extensive robustness analysis of the relationship between trust and growth by investigating a later time period and a bigger sample than in previous studies. They found that the trust–growth relationship is not robust, but is sensitive to sample selection and model specification. Özcan and Bjørnskov (2011) found that trust is significantly associated with faster human development over the 1980–2005 periods and the impact of trust occurs mainly in less democratic countries. Bjørnskov (2012) presented evidence that trust affects schooling and the rule of law directly in a sample of 85 countries, which in turn affect the investment rate and provide a direct effect on the growth rate. Deng, Lin, and Gong (2012) found that social capital can upgrade human capital, improve government efficiency, and thus exert a positive impact on economic growth. Tao et al. (2014) untangled the relationship between social trust and political trust in rural China, with the finding that political trust positively and significantly affects social trust. Superior economic performance and economic modernization increase political trust and indirectly the levels of social trust.

Meanwhile, Temple and Johnson (1998) pointed out: “so many predictions went awry because researchers sought the origins of long-run growth in the wrong places. In particular, they neglected the role of ‘social capability’ in economic development. . . . the broad range of mechanisms that might be relevant.” In the comment, “the broad range of mechanisms” can be understood as institution. Lin (1989) held that institution is a set of behavior rules governing specific actions and relations, and it can be formal or informal. North (1990) viewed institutions as the rules of the game in a society or, more formally, as the humanly devised constraints that shape human interaction. He distinguishes formal from informal rules. Formal institution includes laws, regulations, and contracts signed between economic entities, while the informal institution includes behavior norms, ethics, ideology, and customs.

Thus, social trust can be viewed as a form of informal institution. Although there have been many different definitions of institution, economists agree on the important role of formal institutions in promoting economic growth. For example, North (1981) examined the impact of formal institution and institution change on economic growth and analyzed the influences of such factors as property rights, nationalization, and ideology on economic development. The results showed that institution has a decisive effect on economic growth in western countries.

Since social trust and formal institution are intrinsically related in economy, it is inappropriate to separate them in studies. However, the empirical studies focusing on the relations between social trust, institution, and economic growth are few. Some studies examined their combined effects. For example, Balamoun-Lutz (2005) discovered that social trust has a robust positive influence on income and institution does not have an independent influence, but their interaction has a positive effect in 39 African countries. Farole, Rodriguez-Pose, and Storper (2007) found that community, society, and their interaction are conducive to the efficacy of microeconomic institution, problem-solving institution, and social policy institution across 58 countries, which in turn promote economic growth. Yang and Shen (2010) discovered that structural social capital (e.g., density of nongovernment organization) and institution have significantly positive combined effects, while cognitive social capital (e.g., trust) and institution have insignificant combined effects; some studies examined the effects of social capital

at different levels of formal institution. For example, Ahlerup, Olsson, and Yanagizawa (2007) adopted a nonlinear regression model, where social capital was proxied by social trust and institutions were proxied by government performance index, and argued that the marginal effect of trust on economic growth decreases with institutional strength; some studies examined the impact of trust and formal institution on foreign trade development. For example, Feenstra et al. (2012) demonstrated the importance of institutional quality in China for both the volume of trade and the ability to trade in differentiated goods that rely on contract enforcement. Zhao and Chi (2014) showed that the trust level of host country has obvious positive effects toward China's foreign trade. The institutional difference between China and the host country will increase the transaction cost in international trade and thus will hamper the bilateral trade.

This article puts social trust, formal institution, and economic growth into one framework for theoretical analysis. It examines the relationship between trust and institution¹ and their combined effects on economic growth in China. This article offers some specific contributions to existing literatures. First, different from existing studies using cross-country data (Ahlerup, Olsson, and Yanagizawa 2007; Balamoune-Lutz 2005; Farole, Rodriguez-Pose, and Storper 2007), this article focuses on different regions in China that basically share the same political system, cultural values, and historical traditions, so as to reduce the estimation error caused by possible omitted variables. Second, different from existing studies focusing on the combined effects (Balamoune-Lutz 2005; Farole, Rodriguez-Pose, and Storper 2007), this article explores the relationship between trust and institution and examines the effects of trust on economic growth at different levels of institution. In addition, in order to correct for possible endogenous problems in empirical analysis, this article instruments social trust by using female illiteracy rate and Educated Urban Youths rate depending on Chinese actual situation and conducts robustness tests for the reliability of results.

The results show that the increase in social trust and the development of formal institution both significantly promote economic growth. The effect of social trust on economic growth is closely related to the institution level, and this effect weakens with institutional strength. The rest of article is organized as follows. The first section presents the theoretical backgrounds of the relationship between social trust and formal institution. The following section introduces data selection and variable measurement. The estimation results are reported and analyzed in the next section, followed by robustness tests. The last section provides policy implications and concludes the article.

Theoretical Backgrounds

From an economic point of view, social trust acts itself as a medium of exchange and communication. It is an optimistic expectation where, before finishing a deal, the principal believes that his agent will not make use of the latter's information advantages to harm his interests; or it is an optimistic expectation that the partner will cooperate in transaction under uncertain circumstances. Generally speaking, it is believed that the opportunism owing to information asymmetry leads to high transaction costs, while trust makes both parties constrain opportunistic behavior in pursuit of long-term interests. In this sense, social trust is the lubricant of economic transactions. It not only helps to increase the investment rate and to accumulate physical capital by reducing transaction costs, but also is able to create higher returns on human capital and technology, thus acting as an important factor in promoting economic growth.

Meanwhile, a well-functioning formal institution also contributes to economic growth. From the macroperspective, a better institution is able to reduce uncertainty in transaction by lowering transaction and information cost, and thus to improve the efficacy of production factors. From the micro-perspective, a better institution may help to provide a favorable environment for people to invest in physical and human capital, to diffuse advanced knowledge, as well as to enhance cooperation. By protecting property rights and intellectual property, it is also able to effectively encourage individuals to give full play to their talents and thus to create more value.

Social trust and formal institution are inextricable and intrinsically related in economy. As there are risks and limitations in trusting behavior itself, its reliance on institutions cannot be ignored. To some degree, trust is based on the development of formal institutions. A better institution enables the constraint of bad behaviors and encouragement of good ones, hereby contributing to the forming of social trust. Despite the fact that trust can be generated utterly from interpersonal relationships, it cannot be stable as always without the support of formal institutions. However, only those formal institutions that are trustworthy are conducive to the formation of mutual trust. In addition, even if better institution can be learned from abroad, they will definitely produce conflicts with indigenous informal institutions if the latter resists changing due to inertia, so that the better institution cannot be implemented and thus bring no practical effects.

To a certain extent, social trust and formal institution are substitutable. In a society with higher level of social trust, people's reliance on formal institution is less and the demand for institution is becoming weaker. Social trust is able to compensate for the deficiency of formal institutions, especially when the government is unwilling or unable to provide effective protection for private property or contract enforcement. Meanwhile, a better protection of property rights, an effective control over the power of the privileged people, and a relatively stable social and institutional environment will help people to reach an agreement and compromise, thus reducing the reliance on informal institutions. However, formal institutions can never eliminate the interpersonal risks, thus requiring social trust to serve as a complement. Especially in relatively more complicated and uncertain circumstances, social trust and formal institution can jointly solve many disputes and conflicts in society. Once one factor becomes weak in function, the other is becoming strong.

Regarding the above all, we demonstrate that the increase in social trust significantly promotes economic growth. The improvement of formal institution is also beneficial to economic growth. The effect of social trust on economic growth depends on the institution level. At lower levels of institutions, social trust exerts a positive impact on economic growth. As the levels of institutions get higher, the effect of social trust weakens and the effect of institutions strengthens.

The Data

The dependent variables in this article are growth rates of GDP per capita in different provinces in China. The growth rate is calculated the same way as that used by Barro and Sala-i-Martin (1991) and is denominated to the price of the year 2000. Due to data availability on the main variables such as trust and institution, the data span from the year 2001 to 2009. The data are obtained from *China Statistical Yearbook* and *China Labor Statistical Yearbook*, and are discussed in detail below.

Measuring Social Trust

The first step in our empirical study is to identify a reasonable measure of social trust at Chinese provincial level. Due to the lack of the complete set of statistical data on social trust within China, our trust measure is obtained from WVS, which is recognized as the best and probably only reasonable source for making comparisons of values and perceptions across regions and countries over time. The samples were selected using a combination of probability-proportional-to-size and multistage sampling techniques. The question used to assess the level of trust in a society is “*Generally speaking, would you say that most people can be trusted, or that you can't be too careful in dealing with people?*” Three candidate answers are (i) *Most people can be trusted*, (ii) *Can't be too careful*, and (iii) *Don't know*. Although WVS has been adopted by many cross-country and within country studies (Bjørnskov 2012; Knack and Keefer 1997; Wang and Sheng 2010; Zak and Knack 2001; Zhao and Chi 2014), some scholars have questioned its appropriateness, such as translation difficulty, question and data inconsistency, and differential response bias. But after comparing with independent data source, it has

proved to be a fairly valid measure of honesty, trust, and trustworthiness. For example, Knack and Keefer (1997) discovered that these problems did not introduce noises, but rather captured universal interpersonal trust precisely. Uslaner (2002), Bjornskov (2007), Sapienza, Toldra, and Zingales (2007), and Cox et al. (2009) have also proved the appropriateness of this measure from different perspectives.

By the year of 2010, WVS has comprised four waves in China, separately in 1990, 1995, 2001, and 2007, with the sample size of 1000, 1500, 1000 and 2015, respectively. In 1990, 2001, and 2007 surveys, the home province of each respondent was clearly marked out, while in 1995 only East or West and North or South were marked out. Also, some responses and home provinces of the respondents were missing. As a result, the number of the valid responses was 983, 963, and 1873 respectively, with the sum of 3819. Table A1 provides the tabulation of the trust score by province and year. Due to the limitation of valid responses, our trust variable is measured by the proportion of respondents in each province replying “*Most people can be trusted*” after combining 1990, 2001, and 2007 responses and then deleting the “*don’t know*” responses. This measure is feasible, because the level of trust changes very slowly with time. For example, Yang and Shen (2010) argued that the trust levels of different provinces in China remain almost unchanged during the past several years. Yang, Meng, and Fang (2011) noted that the average social trust level is 0.53 between 1990 and 2010 in China, with the standard deviation of only 0.06. Knack and Keefer (1997) discovered that for 20 market economies with trust levels of the year 1981 and 1990, the correlation between the two is as high as 0.91. Uslaner (2002), Bjornskov (2007), and Tabellini (2008) also discovered that trust remains almost unchanged within a very long period of time, and such a period can last for as long as half a century.

Measuring Institution

Another important variable to measure is formal institution. In cross-country empirical studies, measures of institution include a host of indicators such as property rights (Knack and Keefer 1995; Zak and Knack 2001), bureaucratic structure (Rauch and Evans 2000), political rights, and civil liberties (Isham, Kaufmann, and Pritchett 1997; Kormendi and Meguire 1985; Scully 1988). But within China, these indicators do not exhibit great differences between provinces. Due to data availability and limitation, most studies adopt Fan, Wang and Zhu’s (2011) marketization index to measure institutional quality in China (Deng, Xie, and Li 2014; Shi, Chen, and Shi 2015; Yang and Shen 2010; Zhou 2014). This index covers 23 basic indicators in five subindices, including the government and market relations, the development of non-nationalization economy, the development of product market, the development of factor market, and the development of market intermediaries and rule of law. However, this index focuses on economic institution, especially the development of private sector. This article argues that the two subindices, namely, the government and market relations and rule of law,² are able to capture the quality of institution in a province in the process of reform and opening up in China.

Thus, our measure of institution is obtained by calculating the arithmetic average of all indicators contained in the two subindices,³ see Table A2. In this measurement, the base year is 2001. The level of institution commonly ranges between 0 and 10, but in some circumstances, it may be above 10 or below 0 as the year changes. Generally, the higher the number, the better is the quality of institution. Besides the general measurement, this article also examines the two subindices (the government and market relations, and rule of law) and their effects on economic growth, respectively.

Instrumental Variables

In order to overcome the possible endogenous problems in our empirical analyses, this article adopts two instrumental variables of social trust based on Chinese actual situation: female illiteracy rate and educated urban youths rate.

(1) *Female illiteracy rate*. In reference to Akcomak and TerWeel (2009), the female illiteracy rate is defined as the percentage of illiterate and semi-illiterate female in the female population at and above the age of 12 in the year of 1987. Since the *Nation Bureau of Statistics of China* did not publish corresponding data before 1987, the illiteracy rate in 1987 is the earliest data available. The data are obtained from *China Population Statistics Yearbook of 1988*. The reason why female illiteracy rate rather than the overall population illiteracy rate is used is that there has been a tradition in China that male is preferred to female, and the education in the past thousands of years had been received only by males, especially in rural areas. (2) *Educated urban youths rate*. The Educated Urban Youths started to work in rural and mountain areas in the 1950s, which became a compulsory arrangement since December 1968; it is a part of China's "Ten-Year Cultural Revolution." Liang and Li (2013) argued that this arrangement affects these youths' trust levels through several ways. For example, they lose the opportunity to continue their education; they are frustrated and lack the desire to struggle for better life; they realize the perishing of social moralities and become numbness to some unfair phenomena. They found that the overall trust level of Educated Urban Youths is much lower than the comparable social groups, after controlling for income, education, and other related factors. Zheng (2001) argued that it may also affect the trust level of peasants in rural areas. Because China has a huge population with relatively little arable land, those youths coming to the rural areas were competing with the peasants for land, rations, and working performance, which had increased the burdens of the peasants. This article defines Educated Urban Youths rate as the percentage of Educated Urban Youths in the nonagricultural population during the 1962–1979 period. The data in Hainan and Chongqing are missing, because these two regions were established after 1979. The data on Educated Urban Youth are obtained from Gu (2009) and the data on nonagricultural population are obtained from the *Comprehensive Statistical Data and Materials on 50 Years of New China (1949–1999)*.

Environmental Variables

(1) *Saving rate*. In neoclassical economic growth model, both the stock of capital and the output per unit of effective labor are affected by saving rate in steady state. The higher the saving rate, the higher are the stock of capital and the output per unit of effective labor. Adopting Barro (1991) and Knack and Keefer (1997)'s methods, the saving rate is measured by the average proportion of total capital to GDP, where total capital is the sum of total fixed capital and inventory changes. (2) *Population growth rate*. In neoclassical economic growth model, both the stock of capital and the output per unit of effective labor are affected by population growth rate in steady state. The lower the population growth rate, the higher are the stock of capital and the output per unit of effective labor. The average annual population growth rates are calculated the same way as that of growth rates of GDP per capita. (3) *Initial GDP per capita*. Based on neoclassical economic growth model and economic convergence theory, the growth rate of an economy is closely related to its initial GDP level. The greater its initial GDP is lower than its steady-state level, the higher its growth rate will be; thus, the economies with similar preference and technology will converge to the same state. This article defines initial GDP as the logarithm of GDP per capita in 2001.

Empirical Results

Descriptive Statistics

The descriptive statistics of variables are presented in Table 2. We find that the main variables show great differences between different provinces in China. For instance, the maximum growth rate of GDP per capita is 23.27% and the minimum is only 3.25%. The maximum annual growth rate is 17.21%, which is more than twice of the minimum value, i.e., 7.85%. The average social trust level is 0.55, meaning that about 55% of people believe that most people can be trusted on average, which is

Table 2. Descriptive statistics.

	Obs	Mean	Standard deviation	Maximum	Minimum
GDP growth rate	243	11.34	2.83	23.27	3.25
Social trust	243	0.55	0.10	0.88	0.38
Institution	243	6.66	1.93	9.59	−3.04
Government–market relation	243	7.33	2.44	10.53	−4.66
Rule of law	243	5.51	1.68	9.97	−0.34
Saving rate	243	51.02	11.77	96.7	30.9
Population growth rate	243	0.78	1.24	5.51	−5.71
Initial GDP	243	8.95	0.50	10.31	7.95

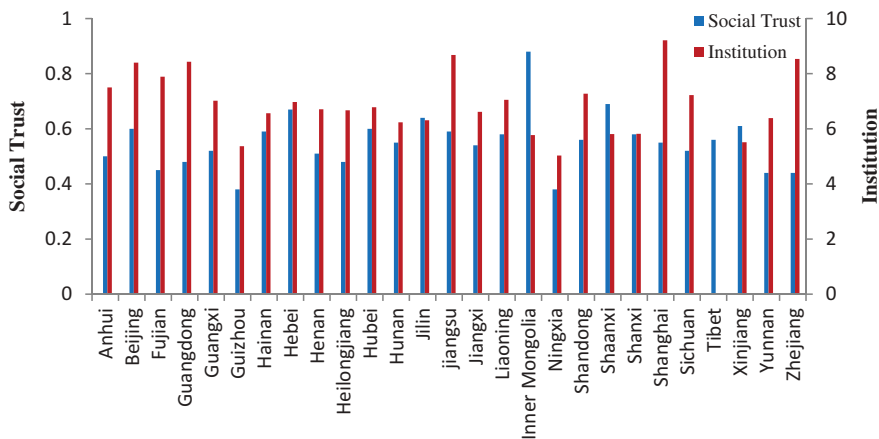


Figure 1. Social trust and institution by province in China.

very close to Yang, Meng and Fang’s (2011) study. The standard deviation is 0.10, and the coefficient of variance used to measure the degree of dispersion is 18.5%, implying that differences between provinces are comparatively large. Among them, Inner Mongolia has the highest social trust of 0.88, while Guizhou and Ningxia have the lowest of only 0.38, which can be seen in Figure 1. With regard to institution, Shanghai, Jiangsu, Zhejiang, Guangdong, and Beijing have the higher institution level of over 9, meaning that their institutional quality are improved and better developed. Tibet has the lowest of only 0.81. Since the level ranges from 0 to 10, there is no doubt that the differences in institutional quality are also remarkable.

Table 3 shows simple correlation coefficients between major variables. It is noteworthy that both social trust and institution are positively correlated with economic growth at the 5% significance level. The simple correlation between social trust and institution is negative, but not significant. But after controlling for other variables, the partial correlation coefficient is -0.2751 and significant at 1% significant level. As can be seen in Figure 1, the institution level of Inner Mongolia, with the highest social trust, is 5.17, and that of Shaanxi, Hebei, Jilin, and Xinjiang, with the relatively higher social trust, is 5.39, both of which are lower than the national average institution level of 6.15. Likewise, although Zhejiang, Shanghai, Guangdong, and Jiangsu have established comparably well-functioning institution, their average social trust levels are lower than the national average of 0.51.

Table 3. Simple correlations.

	GDP growth rate	Social trust	Institution	Saving rate	Population growth rate	Initial GDP
GDP growth rate	1.0000					
Social trust	0.2628*** (0.0000)	1.0000				
Institution	0.1257** (0.0504)	−0.0727 (0.2590)	1.0000			
Saving rate	0.2252*** (0.0004)	0.0988 (0.1245)	−0.2937*** (0.0000)	1.0000		
Population growth rate	−0.5512*** (0.0000)	0.0282 (0.6614)	0.1266** (0.0486)	0.0781 (0.2250)	1.0000	
Initial GDP	−0.2363*** (0.0002)	0.1153* (0.0729)	0.5434*** (0.0000)	−0.2353*** (0.0002)	0.5472*** (0.0000)	1.0000

Note: *p*-values are in parentheses. ***, **, and * indicate statistical significance at 1%, 5%, and 10%, respectively.

Direct Effects of Social Trust and Institution

Due to data availability and limitation, the data span 27 provinces in the period 2001–2009.⁴ Since in our panel dataset, the main variables social trust and initial GDP per capita are both invariant with time, so fixed effect estimations cannot be employed. Thus, a series of pooled ordinary least square (OLS) estimations and random-effect estimations are performed in empirical analysis. The *p*-values of Breusch–Pagan Lagrangian multipliers are all less than 0.1; thus, random-effect estimations are preferred.

Table 4 first presents estimates of direct effects of social trust on per capital GDP growth, after controlling for saving rate, population growth rate, and initial GDP. As can be seen, whether employing random-effect estimation or OLS estimation, the coefficients of social trust are both significant and have little difference. In column (1), the coefficient of social trust is 6.380, meaning that if the level of social trust increases by 1 unit, the growth rate increases by 6.380% accordingly; or if social trust changes by one unit of standard deviation, the growth rate changes by 0.255 units of standard deviation accordingly. In economic sense, if social trust levels of Guizhou and Ningxia rise from the lowest level of 0.38 to the average, their GDP growth rates in 2009 will increase by 1.09%, from 13.26% and 10.61% to 14.35% and 11.7% respectively.

The next six columns present estimates of direct effects of institution on per capital GDP growth. The coefficients of institution are all significantly positive. The coefficient in column (3) is 0.695, meaning that the 1 unit increase in the level of institution raises growth rate by about 0.628%, or if the level of institution changes by one unit of standard deviation, the growth rate changes by 0.474 units of standard deviation, which is about 1.341%. A well-functioning institution promotes economic growth dramatically. Specifically, the two subindices of institution, government–market relation and rule of law, also have positive effects, with the coefficients of 0.565 and 0.514, respectively. This demonstrates that correctly locating the function of the government and actively giving play to the markets’ fundamental role in resources allocation are indispensable in the development of formal institution and economic growth.

In Table 4, saving rate exhibits a positive and significant effect on growth. In column (1), the coefficient of saving rate is 0.085, implying that a 1% rise in saving rate is associated with an increase in growth of 0.085%. Population growth rate exhibits a negative and significant effect on growth. The coefficient is −1.420, implying that a 1% rise in population growth rate is associated with a decrease in growth of 1.420%. These results are consistent with our previous analysis. According to neoclassical economic growth model, the higher the saving rate, the higher the output per unit of effective labor in

Table 4. Direct effects of social trust and institution on economic growth.

	(1) RE	(2) OLS	(3) RE	(4) RE	(5) RE	(6) OLS	(7) OLS	(8) OLS
Social trust	6.380*** (2.055)	6.521*** (1.363)						
Institution			0.695*** (0.103)			0.430*** (0.087)		
Government–market relation				0.565*** (0.085)			0.279*** (0.065)	
Rule of law					0.514*** (0.125)			0.520*** (0.122)
Saving rate	0.085*** (0.014)	0.071*** (0.012)	0.079*** (0.013)	0.089*** (0.013)	0.064*** (0.015)	0.089*** (0.012)	0.093*** (0.013)	0.067*** (0.013)
Population growth rate	−1.420*** (0.142)	−1.544*** (0.134)	−1.252*** (0.135)	−1.255*** (0.136)	−1.321*** (0.142)	−1.458*** (0.136)	−1.467*** (0.138)	−1.519*** (0.136)
Initial GDP	0.907** (0.463)	1.000*** (0.343)	−0.650 (0.588)	−0.212 (0.577)	−0.442 (0.618)	0.238 (0.396)	0.637* (0.372)	−0.177 (0.481)
Constant	−3.497 (4.290)	−3.641 (3.239)	9.502* (5.160)	5.519 (5.116)	10.211* (5.443)	2.933 (3.426)	−0.010 (3.327)	7.840* (4.113)
N	243	243	243	243	243	243	243	243
R-squared	0.4539	0.4598	0.4387	0.4091	0.4478	0.4634	0.4504	0.4498
F		50.64 [0.0000]				51.38 [0.0000]	48.76 [0.0000]	48.65 [0.0000]
Wald χ^2	151.89 [0.0000]		199.07 [0.0000]	196.50 [0.0000]	159.03 [0.0000]			

Note: (1) The dependent variables are GDP growth rates of different provinces in China. (2) Standard errors are in parentheses. (3) ***, **, and * indicate statistical significance at 1%, 5%, and 10%, respectively. (4) *p*-values of Breusch–Pagan Lagrangian multipliers are 0.0001, 0.0000, 0.0000, and 0.0000 in columns (1), (3), (4), and (5), respectively. Null hypothesis is that no significant difference exists across units, i.e., no panel effect.

steady state, and therefore the faster the growth will be; the higher the population growth, the lower the output per unit of effective labor, and therefore the lower the growth will be.

It is interesting to mention that the coefficients of initial GDP are only negative in columns (3), (4), (5), and (7), but not significant, implying that the regional economic convergence is not evident in China on the whole.⁵ This echoes with some existing studies. For example, De Long (1988) found no evidence of convergence by sampling not only industrial countries, but also non-industrial countries. Cai and Du (2000) found that convergence exists within each region, that is, the East, Center, and west of China, but differences between regions continue to enlarge. Qin (2004) reported that the difference in income per capita across regions has constantly enlarged from 1990 to 1999, and the growth rate is significantly higher in the east than that in the west of China.

Joint Effects of Social Trust and Institution

The previous section shows that social trust is conducive to promoting economic growth. Now the question is what are the relations between social trust, as a form of informal institution, and formal institution? What are their combined effects? Now we add the interaction term between institution and trust into our regression to investigate their combined effects on economic growth.

In the first six columns of Table 5, the coefficients of social trust are all significantly positive. The coefficient is 8.387 in column (1), implying that growth rate will rise by 8.378% if social trust rises by 1 unit. In comparison with the coefficients in Table 4, the effect of social trust has dramatically increased. A plausible explanation is that trust and institution are negatively correlated, and the variable social trust in Table 4 has already captured the effect of institution. Thus, after controlling for institution, the coefficients of social trust have increased. Meanwhile, the positive effects of institutions are also nonnegligible, even when considering the variable social trust. The coefficient of institution is 0.690, implying that the 1 unit increase in the level of institution raises the growth rate by about 0.690%. Specifically, the effects of government–market relation and rule of law both remain significantly positive, with the coefficients of 0.549 and 0.536, respectively; thus it is very reasonable to foster well-functioning market intermediaries, strengthen the protection of producers and consumers' rights, and build a better legal environment, so as to promote regional economic growth. After adding the interaction term between trust and institution, in columns (10)–(12), the coefficients of social trust and institution are both significantly positive and the interaction terms are significantly negative. It indicates that the effects of social trust on economic growth depend on the level of institution. But in columns (7)–(9), the coefficients are not significant any more, indicating that their combined effects are rather weak. In order to examine the relations between social trust, institution, and economic growth precisely, this article examines the effects of social trust at different levels of institution in the next section.

The Effect of Social Trust at Different Levels of Institution

To dig deeper into the relations between social trust, institution, and economic growth, this article divides all the sample provinces into three groups based on the average level of institution, and then makes regressions for each group.⁶ The three groups remain unchanged during the sample period, because it is easy to see that such division is rather similar to the east–center–west division of China. Generally, the east of China has better developed and more complete institution followed by the central, and the west has the weakest quality of institution. These differences in institution can be traced back to their historical experiences, their specific development paths, as well as the incremental reform and opening-up policy in China.⁷

Table 6 reports estimation results for each group. As can be seen, the effects of social trust on economic growth depend on the level of institution. The coefficients of social trust are significantly positive only in regions of low institution levels, with its coefficients reaching as high as 13.8. The

Table 5. Joint effects of social trust and institution on economic growth.

	(1) RE	(2) RE	(3) RE	(4) OLS	(5) OLS	(6) OLS	(7) RE	(8) RE	(9) RE	(10) OLS	(11) OLS	(12) OLS
Social trust	8.387*** (2.092)	8.728*** (2.174)	6.807*** (1.940)	7.794*** (1.283)	7.833*** (1.305)	6.772*** (1.308)	10.784 (7.223)	10.836 (7.597)	10.363* (6.098)	20.365*** (6.220)	22.489*** (6.443)	12.529** (5.488)
Institution	0.690*** (0.097)			0.509*** (0.082)			0.888 (0.634)			1.649*** (0.558)		
Government–market relation		0.549*** (0.079)			0.345*** (0.062)			0.697 (0.614)			1.567*** (0.530)	
Rule of law			0.536*** (0.121)			0.545*** (0.116)			0.900 (0.603)			1.137** (0.560)
Saving rate	0.076*** (0.013)	0.087*** (0.013)	0.059*** (0.014)	0.081*** (0.012)	0.086*** (0.012)	0.057*** (0.012)	0.078*** (0.013)	0.089*** (0.013)	0.061*** (0.015)	0.085*** (0.012)	0.091*** (0.012)	0.061*** (0.013)
Population growth rate	−1.249*** (0.132)	−1.247*** (0.133)	−1.336*** (0.138)	−1.373*** (0.128)	−1.377*** (0.130)	−1.463*** (0.130)	−1.256*** (0.132)	−1.260*** (0.133)	−1.328*** (0.139)	−1.354*** (0.127)	−1.359*** (0.129)	−1.445*** (0.131)
Initial GDP	−0.849 (0.519)	−0.405 (0.506)	−0.665 (0.564)	−0.268 (0.378)	0.169 (0.356)	−0.520 (0.462)	−0.775 (0.492)	−0.295 (0.467)	−0.671 (0.561)	−0.309 (0.376)	0.129 (0.353)	−0.545 (0.462)
Social trust* Institution							−0.408 (1.142)			−2.059** (0.997)		
Social trust*								−0.347 (1.100)	−2.194** (0.945)			−1.117 (1.034)
Government–market relation												
Social trust* Rule of law									−0.690 (1.123)			
Constant	6.793 (4.468)	2.647 (4.439)	8.639* (4.901)	3.010 (3.193)	−0.328 (3.107)	7.491* (3.907)	4.892 (5.680)	0.739 (5.753)	6.699 (5.786)	−3.851 (4.593)	−8.471* (4.666)	4.456 (4.811)
N	243	243	243	243	243	243	243	243	243	243	243	243
R-squared	0.5258	0.5043	0.5043	0.5357	0.5229	0.5057	0.5314	0.5141	0.5065	0.5439	0.5335	0.5082
F				54.68 [0.0000]	51.95 [0.0000]	48.50 [0.0000]				46.91 [0.0000]	44.99 [0.0000]	40.64 [0.0000]
Wald χ^2	226.10 [0.0000]	220.44 [0.0000]	185.19 [0.0000]				229.39 [0.0000]	222.84 [0.0000]	186.25 [0.0000]			

Note: (1) The dependent variables are GDP growth rates of different provinces in China. (2) Standard errors are in parentheses. (3) ***, **, and * indicate statistical significance at 1%, 5%, and 10%, respectively. (4) *p*-values of Breusch–Pagan Lagrangian multipliers are 0.0000, 0.0000, 0.0002, 0.0000, 0.0000, and 0.0004 in columns (1), (2), (3), (7), (8), and (9), respectively. Null hypothesis is that no significant difference exists across units, i.e., no panel effect.

Table 6. Social trust, institution, and economic growth at different levels of institution.

	High		Medium		Low	
	(1) RE	(2) OLS	(3) RE	(4) OLS	(5) RE	(6) OLS
Social trust	−1.212 (4.839)	−0.244 (4.140)	2.157 (4.675)	2.214 (2.719)	13.797*** (2.225)	13.793*** (2.210)
Institution	0.769*** (0.216)	0.789*** (0.213)	1.153*** (0.204)	1.177*** (0.204)	0.315** (0.138)	0.312** (0.137)
Saving rate	0.089** (0.042)	0.072* (0.039)	0.013 (0.026)	0.011 (0.025)	0.092*** (0.019)	0.092*** (0.019)
Population growth rate	−1.004*** (0.254)	−1.122*** (0.241)	−1.358*** (0.198)	−1.402*** (0.200)	−1.504*** (0.259)	−1.508*** (0.259)
Initial GDP	−1.259* (0.727)	−1.000 (0.667)	−0.184 (1.007)	−0.217 (0.593)	−3.014** (1.245)	−3.004** (1.238)
Constant	14.312** (6.541)	12.158** (5.921)	3.780 (8.104)	3.996 (4.883)	23.658** (9.871)	23.584** (9.817)
N	81	81	81	81	81	81
R-squared	0.5425	0.5444	0.6462	0.6464	0.6079	0.6079
F		17.93 [0.0000]		27.42 [0.0000]		23.26 [0.0000]
Wald χ^2	73.60 [0.0000]		134.02 [0.0000]		115.44 [0.0000]	

Note: (1) The dependent variables are GDP growth rates of different provinces in China. (2) Standard errors are in parentheses. (3) ***, **, and * indicate statistical significance at 1%, 5%, and 10%, respectively. (4) *p*-values of Breusch–Pagan Lagrangian multipliers are 0.4507, 0.5780, and 0.3933 in columns (1), (3), and (5), respectively. Null hypothesis is that no significant difference exists across units, i.e., no panel effect.

positive role of social trust, as a form of informal institution, is brought into full play. As the level of institution becomes higher, the effects of social trust weaken. In regions of medium and high institution levels, the coefficients are not significant anymore. The other side of the coin is that in these regions, the coefficients of institution are all significantly positive. The positive effects of institution appear most dramatic at medium institution level. On average, the increase in institution by 1 unit will bring about 1.16% economic growths. While in regions of low institution levels, the effects of institution decrease dramatically, in accordance with our previous analysis.

Robustness

For the reliability of our results, this article conducts robustness analyses as follows. First, the above estimations may have possible problems of endogeneity. On the one hand, if social trust stems from optimistic sentiments brought by an increase in GDP and income level, the causal relationship may exist between trust and growth. On the other hand, a region with higher social trust level can more easily carry out economic stimulus policies in an effective manner. So it is likely that the variable social trust is correlated with the possible omitted variables that are captured in the error term, if we fail to incorporate variables to measure such policy performances into the estimation. In order to correct for such endogenous problems, this article uses female illiteracy rate and Educated Urban Youths rate as instruments for social trust.

Both 2SLS and 3SLS estimations are conducted on economic growth. Due to space limitation, Table 7 reports 2SLS estimation results only with the results of 3SLS consistent with 2SLS.⁸ The first-stage estimates suggest that both instruments are significantly correlated with trust. *F*-tests for the joint significance of the instruments and minimum eigenvalue statistics always exceed the critical value of

Table 7. Social trust, institution, and economic growth: IV estimation.

	(1) Full sample	(2) Full sample	(3) High	(4) High	(5) Medium	(6) Medium	(7) Low	(8) Low
Social trust	4.787** (2.024)	6221*** (1.917)	5.560 (10.771)	8.009 (10.284)	-2.239 (5.640)	-1.810 (4.084)	9.791*** (3.167)	9.434*** (3.082)
Institution		0.532*** (0.084)		0.593*** (0.196)		1.341*** (0.213)		0.391** (0.172)
Saving rate	0.073*** (0.013)	0.075*** (0.011)	0.072 (0.055)	0.080 (0.049)	0.111*** (0.022)	-0.026 (0.027)	0.086*** (0.019)	0.081*** (0.018)
Population growth rate	-1.550*** (0.135)	-1.383*** (0.126)	-1.064*** (0.289)	-1.130*** (0.275)	-1.691*** (0.227)	-1.478*** (0.177)	-1.803*** (0.245)	-1.519*** (0.273)
Initial GDP	1.047*** (0.349)	-0.760* (0.441)	-0.453 (0.717)	-1.475** (0.712)	0.938 (0.828)	-1.293** (0.595)	-0.967 (1.414)	-1.781 (1.463)
Constant	-3.153 (3.324)	8.662** (3.468)	10.329 (7.761)	13.450** (6.687)	-0.389 (5.884)	17.522*** (4.813)	10.490 (10.928)	16.034 (11.158)
N	234	234	81	81	72	72	81	81
R-squared	0.4639	0.5346	0.4359	0.5094	0.5025	0.7138	0.5658	0.5893
Wald χ^2	185.98	258.05	66.08	86.83	75.12	179.24	81.20	89.06
Sargan χ^2	[0.0000] 0.081 [0.7764]	[0.0000] 2.633 [0.1047]	[0.0000] 1.818 [0.1776]	[0.0000] 0.859 [0.3540]	[0.0000] 8.555 [0.0034]	[0.0000] 5.471 [0.9885]	[0.0000] 0.019 [0.8897]	[0.0000] 8.601 [0.0034]

Note: (1) The dependent variables are GDP growth rates of different provinces in China. (2) Standard errors are in parentheses. (3) ***, **, and * indicate statistical significance at 1%, 5% and 10%, respectively. (4) Sargan is a test of over identification. Null hypothesis is that over-identifying restrictions are valid.

10, indicating that female illiteracy rate and Educated Urban Youths rate are not weak instruments. Also the null hypothesis that the over-identifying restrictions are valid is never rejected. In full-sample estimations, both social trust and institution exert positive effects on economic growth, and their coefficients are 6.221 and 0.532, respectively. The effects of saving rate are significantly positive, and the effects of population growth rate are significantly negative, which coincide with our previous results. The last six columns present the estimates of the effects of social trust at different levels of institution. As can be seen, the effects of social trust weaken with institutional strength. Social trust matters the most when institutions are weak and almost ceases to matter when institutions get strong. Thus, our previous results are robust.

Second, this article adopts a different measure for the main variable institution. Based on all indicators contained in two subindices, namely the government–market relations and rule of law, principal component analysis is employed to obtain the measure for institution, where the weight is the coefficient of the first principal component corresponding to each indicator. It is found that the measures of institutions calculated by the two methods (i.e., arithmetic average and principal component analysis) are very close in number. The main results remain unchanged.

Third, this article reestimates the relations by including additional control variables as follows. (1) *Human capital*. Human capital has been the important explanatory variable in economic growth estimation, although some scholars (Cai and Du 2000; Huang et al. 2013) pointed out that the efficacy of human capital, especially of advanced human capital, is still low in current China; thus, its contribution to economic growth cannot compete with traditional factors such as physical capital and labor force. This article adopts the average education of the year 2001 as the proxy variable for human capital, because Barro (1991) argued that using human capital stock, rather than human capital flow, is a better choice. (2) *Nationalization level*. According to Chen and Feng (2000)'s study, the development of privately owned enterprises can exert positive influences on economy in China. This article defines nationalization level as the percentage of the number of the employees in state-owned enterprises to the number of all employees in economy. (3) *Degree of openness*. In the process of the reform and opening up, different regions exhibit great differences within China. For example, the eastern and coastal regions were granted the favorable policies and opening-up opportunities in the 1980s, but similar policies and opportunities came to the central and western regions only after 1992. Past empirical studies have shown that the degree of openness has a notable impact on economic growth no matter what measurement of openness is used, whether by the proportion of direct foreign investment to GDP or by the proportion of total foreign trade to GDP. This article defines the degree of openness as the proportion of the volume of foreign trade, which is import plus export, to GDP, where the average annual exchange rate of RMB against US dollars is used. The results show that adding the above control variables does not change the basic results of previous estimations.

Fourth, all the regressions are run on different samples. For example, it can be found that the institution level of Tibet is only 0.81, far below the average level of other provinces. In order to exclude the influences of extreme data, this article reexamines the effects of trust and institution on economic growth after deleting these influential observations, and the results are robust.

Conclusions

The purpose of this article is to explore the relationship between social trust, institution, and economic growth. We find that the higher level of social trust is conducive to economic growth. A one standard deviation increase in trust is associated with an increase in growth of 0.225 units of standard deviation, which is 0.638 percentage points. The development of institution is also beneficial to economic growth. The effect of social trust depends on the quality of institution, and this effect decreases with institutional strength.

This article carries some implications for policy. It provides new insights into the study of economic growth in China. In addition to increasing the investment of physical capital and human capital, efforts should also be made to enhance the development of informal and formal institution. Take regional

economic growth for example. The west of China should focus on the development of informal institution, increasing its trust level and social capital. Only by doing so can it gradually narrow its development gap with the east. The central region should take active measures to maintain its trust advantage, reinforce, and strengthen its institutional quality.

It should be noted that further research is needed in at least two directions. First, the more effective measurements of trust and institution should be searched at both national and regional levels. Our measure of trust is obtained from WVS. Although this measure is generally accepted both internationally and domestically, some data are missing and thus the sample is not large enough. Our measure of institution is calculated based on all the indicators contained in the two subindices compiled by Fan, Wang, and Zhu (2011), which put some particular emphasis on economic institution, ignoring other aspects of formal institution, such as bureaucratic structure, political rights, and civil liberties.

Second, this article claims that both social trust and better developed institution contribute significantly to economic growth, but it does not further explore the channels through which trust and institution impact on economic growth. In theory, the potential transmission mechanisms may include reducing transaction costs, increasing the accumulation of physical and human capital, and enhancing technology innovation. But in empirical studies, different studies have reached different results. For example, Akcomak and TerWeel (2009) found that technology innovation is the channel through which social capital promotes economic growth. Bjørnskov (2012) argued that trust affects schooling and rule of law directly, thereby raising economic growth. Deng, Lin, and Gong (2012) discovered that social capital can promote economic development by improving human capital and government performance. On account of these, this article argues that the transmission mechanisms are varied across regions and over time, due to changes in social environments. Since the reform and opening-up policy, China has undergone significant changes in economic and social environments. Thus, the mechanisms may not be stable. Therefore, the future research could focus on the possible channels through which social trust and institution promote economic growth, and this would be challenging.

Notes

1. For simplicity, institution is intended to refer to formal institution in the following unless otherwise specified.

2. The first subindex, the government and market relations, includes five basic indicators, which are market share of economic resources distribution, reduction of the farmers tax and fee burdens, reduction of the governments' involvement in business, reduction of enterprises' nontax payments, and reduction of governments' scale. The second subindex, rule of law, includes four basic indicators, which are the development of market intermediaries, protection of producers' rights, protection of intellectual property rights, and the protection of consumers' rights.

3. To obtain the measure of institution, principal component analysis is also used based on all indicators contained in the two subindices. The measures of institution calculated by the two methods (that is, arithmetic average and principal component analysis) are very close in number. However, the problem with principal component analysis is that the weights of the indicators change with data and time. Thus, the levels of institution are not comparable across years, leading to difficulties in analysis.

4. Although there are 31 provinces in China, the trust data are missing in Tianjin, Qinghai, Gansu, and Chongqing. The latest institution data published by Fan, Wang, and Zhu (2011) are in the year 2009, and no corresponding data are published thereafter. Thus, our sample covers 27 provinces from the year 2001 to 2009.

5. As is shown in Table 3, the correlation coefficient between initial GDP and institution is 0.5434, and between initial GDP and population is 0.5472. To check the multicollinearity, we find that in the estimations of (2), (6), (7), and (8), the F values are all close to 50, and variance inflating factors (vif) are 1.34, 1.63, 1.51, and 2.05, respectively. Thus, the models do not incur the problem of multicollinearity. In addition, this article has employed stepwise regressions and found that adding or deleting the variable initial GDP does not affect the significance of main explanatory variables. Since initial GDP is one of the important control variables in explaining economic growth, we keep this variable in the regressions.

6. The first group includes nine provinces of higher institution level, namely Anhui, Beijing, Fujian, Guangdong, Jiangsu, Shandong, Shanghai, Sichuan, and Zhejiang, with the average institution level of 8.128. The second group includes nine provinces of medium institution level, namely Guangxi, Hainan, Hebei, Henan,

Heilongjiang, Hubei, Jiangxi, Liaoning, and Yunnan, with the average level of 6.753. The third group includes the other nine provinces of lower institution level, namely Ghuizhou, Hunan, Jilin, Inner Mongolia, Ningxia, Shaanxi, Shanxi, Tibet, and Xinjiang, with the average level of 5.097.

7. For example, under the past planned economy in China, the low prices of energy and agricultural products are serious impediments to the development of the central and western regions, which widens their gap with the eastern region. In contrast, the eastern region has made great economic progresses due to preferential investment and financial policies, leading to the overall development of the Chinese economy. The competition mechanism of the survival of the fittest makes the eastern regions stronger and the western weaker, causing the Matthew Effect on regional development.

8. In Table 7, 2SLS estimations are conducted in a pooled OLS framework, as it is more commonly used. To check the robustness, the G2SLS estimations are conducted in a panel data framework. The results are robust.

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Table A1. Provinces and social trust data in China.

Panel A: Social trust by province													
	Province	1990	2001	2007	Average	Valid response	Province	1990	2001	2007	Average	Valid response	
1	Anhui	N.A	0.62	0.45	0.50	148	15	Jiangxi	0.62	0.56	0.44	0.54	147
2	Beijing	0.62	0.61	0.58	0.60	187	16	Liaoning	0.68	0.70	0.48	0.58	241
3	Fujian	0.52	0.40	0.38	0.45	190	17	Inner Mongolia	N.A	0.88	N.A	0.88	25
4	Guangdong	N.A	0.54	0.45	0.48	146	18	Ningxia	N.A	N.A	0.38	0.38	24
5	Guangxi	N.A	0.52	0.52	0.52	109	19	Shandong	N.A	0.55	0.57	0.56	349
6	Guizhou	N.A	0.40	0.36	0.38	134	20	Shaanxi	0.58	0.57	0.58	0.58	206
7	Hainan	N.A	N.A	0.59	0.59	61	21	Shanxi	N.A	0.76	0.67	0.69	103
8	Hebei	N.A	0.68	0.66	0.67	131	22	Shanghai	0.58	0.36	0.58	0.55	190
9	Henan	N.A	0.51	0.50	0.51	172	23	Sichuan	N.A	0.52	N.A	0.52	25
10	Heilongjiang	N.A	0.24	0.53	0.48	149	24	Tibet	N.A	0.56	N.A	0.56	25
11	Hubei	0.58	0.62	0.62	0.60	273	25	Xinjing	0.67	N.A	0.53	0.61	109
12	Hunan	N.A	0.55	0.55	0.55	87	26	Yunnan	0.49	0.17	0.49	0.44	169
13	Jilin	0.65	0.63	N.A	0.64	123	27	Zhejiang	N.A	0.72	0.35	0.44	104
14	Jiangsu	0.65	0.48	0.71	0.59	192							
Panel B: Time distribution of valid response													
Year	Sample size	Missing response		Missing response on home province		Valid response		Average trust level					
1990	1000	15		2		983		0.60					
2001	1000	37		0		963		0.55					
2007	2015	142		0		1873		0.52					
Total	4015	194		2		3819		0.55					
Source: www.worldvaluessurvey.org .													

Table A2. Provinces and institution data in China.

	Province	2001	2002	2003	2004	2005	2006	2007	2008	2009	Average
1	Anhui	5.29	5.54	6.03	7.05	8.52	8.75	8.80	8.78	8.78	7.50
2	Beijing	7.65	7.91	7.99	8.69	8.66	8.60	8.60	8.72	8.77	8.40
3	Fujian	6.61	6.87	7.27	7.89	8.37	8.35	8.43	8.54	8.70	7.89
4	Guangdong	7.29	7.39	7.72	8.43	9.11	9.05	9.00	8.99	8.90	8.43
5	Guangxi	6.08	6.43	6.57	6.88	7.33	7.25	7.42	7.64	7.58	7.02
6	Guizhou	4.52	4.54	4.69	5.13	5.74	5.70	5.80	6.16	6.03	5.37
7	Hainan	6.85	6.53	6.23	6.57	6.56	6.58	6.63	6.44	6.71	6.57
8	Hebei	5.46	5.90	6.24	7.09	7.59	7.60	7.55	7.74	7.58	6.97
9	Henan	4.72	5.25	5.74	6.51	7.50	7.51	7.59	7.71	7.88	6.71
10	Heilongjiang	4.24	5.01	5.70	6.79	7.66	7.51	7.64	7.83	7.65	6.67
11	Hubei	4.34	5.12	5.96	6.93	7.59	7.56	7.74	7.90	7.91	6.78
12	Hunan	4.49	5.01	5.49	6.24	7.00	6.81	6.77	6.97	7.36	6.24
13	Jilin	4.60	4.96	5.20	6.10	6.88	6.93	7.16	7.46	7.49	6.31
14	Jiangsu	7.02	7.45	7.80	8.51	9.44	9.41	9.44	9.49	9.57	8.68
15	Jiangxi	4.63	5.02	5.44	6.09	7.37	7.39	7.68	7.90	8.02	6.61
16	Liaoning	5.34	5.64	6.17	6.93	7.80	7.77	7.91	7.99	7.91	7.05
17	Inner Mongolia	4.13	4.86	5.16	5.85	6.28	6.35	6.34	6.44	6.53	5.77
18	Ningxia	3.51	3.90	4.49	5.25	5.54	5.50	5.57	5.84	5.70	5.03
19	Shandong	5.50	5.96	6.39	7.11	8.07	8.07	8.07	8.10	8.23	7.28
20	Shaanxi	4.17	4.63	5.15	5.91	6.32	6.34	6.49	6.63	6.60	5.81
21	Shanxi	4.42	5.05	5.40	6.08	6.21	6.13	6.26	6.49	6.35	5.82
22	Shanghai	8.48	8.72	9.08	9.53	9.41	9.30	9.40	9.59	9.41	9.21
23	Sichuan	5.60	5.89	6.32	7.09	8.14	8.08	8.08	7.99	7.84	7.22
24	Tibet	3.36	0.50	0.47	0.77	-1.10	-0.47	0.06	-0.43	-3.04	0.01
25	Xinjiang	4.50	4.88	5.27	5.88	5.89	5.82	5.75	5.93	5.67	5.51
26	Yunnan	5.01	5.13	5.26	5.63	6.96	7.10	7.43	7.53	7.48	6.39
27	Zhejiang	6.91	7.36	7.78	8.46	9.32	9.19	9.20	9.30	9.29	8.53

Source: Fan, Wang, and Zhu (2011).